

Do Attention Blocks Help Small Medical Imaging Datasets? A Negative Result

Demonstration research paper - synthetic study, fictional data

Abstract

On three small medical imaging datasets (1,200 to 4,800 images), adding attention blocks to CNN backbones produced no statistically significant improvement in AUC and increased overfitting. We find that attention mechanisms do not improve accuracy in low-data regimes and can degrade calibration.

Methods

We replicated five published attention architectures under a fixed training budget with nested cross-validation. Calibration was measured with expected calibration error.

Results

Mean AUC delta across datasets was +0.002 ($p = 0.61$). Expected calibration error worsened from 0.041 to 0.058 with attention enabled. Training time increased 31 percent.

Discussion

Benefits reported on large datasets may not transfer to small clinical cohorts. We recommend regularization-first baselines before architectural additions.